

Name: _____

KILLARA HIGH SCHOOL



HSC MATHEMATICS ASSESSMENT

TERM 2, 2003

Time Allowed: 70 minutes

Maximum Marks: 50 marks

- Instructions:**
- * approved calculators may be used.
 - * all necessary working must be shown.
 - * start each question on a new page.
 - * use one side of page only.
 - * answer in blue or black biro, pencil for graphs and diagrams.

FORMAT:

Question 1 - Geometry - Coordinate and Plane	14 marks
Question 2 - Trigonometric Functions	14 marks
Question 3 - Exponential and Logarithmic Functions	<u>22 marks</u>
TOTAL:	<u>50 marks</u>

Question 1 Geometry (14 marks)

- (a) (i) Neatly draw a number plane, marking the following points:

1

O, the origin

M, (5, 0)

N, (8, 4)

P, (0, 10)

- (ii) Show that the equation of MN is $4x - 3y - 20 = 0$

2

- (iii) Find the gradient of NP.

1

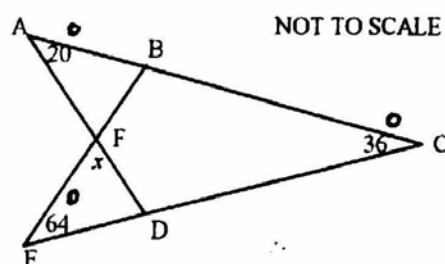
- (iv) Show that MN and NP are perpendicular.

2

- (v) Find the length of MN.

2

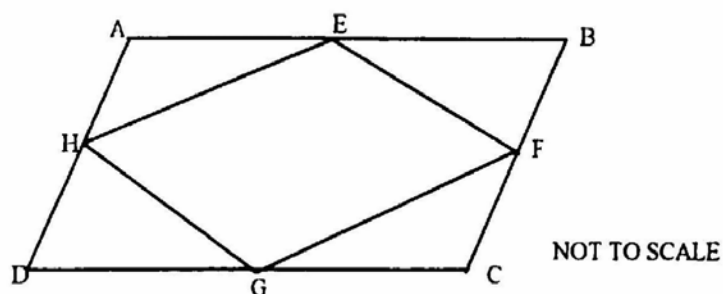
- (b)



2

Find the value of x , giving all reasons.

- (c) In the diagram, ABCD is a parallelogram. Points E, F, G and H are the mid points of AB, BC, CD and DA respectively.



- (i) Prove that $\triangle AEH$ is congruent to $\triangle FGC$

3

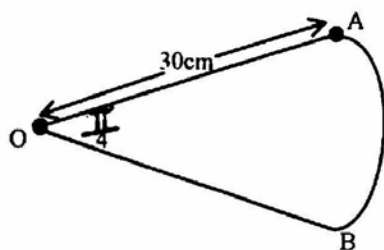
- (ii) What other information is needed to show that EFGH is a parallelogram?

1

(a) (i) Write $53^\circ 45'$ in radians in terms of π . 1

(ii) Find to 4 decimal places $\sec \frac{3\pi}{4}$. 1

(b)



NOT TO SCALE

In the diagram, AB is the arc of a circle with centre O. The radius OA is 30cm. Angle AOB is $\frac{\pi}{4}$ radians.

Find the exact area of sector AOB. 2

(c) Differentiate with respect to x :

(i) $\tan 6x$ 1

(ii) $\frac{x^3}{\sin x}$ 1

(iii) $\frac{1}{\cos x}$ 1

(d) Integrate:

(i) $\sin 2x \, dx$ 1

(ii) $\int_0^\pi (\cos \theta - 1) d\theta$ 1

(e) (i) Neatly draw the graph of $y = 3 \cos 2x$ in the domain $0 \leq x \leq \pi$ 2

(ii) Find the area bounded by the curve $y = 3 \cos 2x$ and the x axis between $x = 0$ and $x = \frac{3\pi}{4}$. Give your answer correct to 3 decimal places. 3

- (a) (i) Write as a fraction in simplest form $125^{-\frac{2}{3}}$ 1
- (ii) Simplify $4^{3x} \times 2^{4x} \div 8^{2x}$ 1
- (b) Express as a numeral $\log_5 50 + \log_5 10 - \log_5 4$ 1
Give your answer correct to 3 decimal places.
- (c) (i) If $\log_6 2 = 0.43$ and $\log_6 3 = 0.68$, evaluate $\log_6 48$ correct to 2 decimal places. 1
- (ii) Find the value of $\log_9 14$ correct to 3 decimal places. 1
- (d) On the same number plane, graph: 3
- (i) $y = 3^x$
- (ii) $y = \log_3 x$
- (e) Differentiate:
- (i) e^{-2x} 1
- (ii) $\frac{e^x}{e}$ 1
- (iii) $\log_e(5x - 4)$ 1
- (iv) $\frac{\log e^x}{3x + 1}$ 1
- (f) Integrate:
- (i) $\int e^{2x}$ 1
- (ii) $\int \frac{4x}{2x^2 + 3}$ 1
- (g) Find $\int_2^5 \frac{x^2}{x^3 - 1} dx$ giving your answer correct to 2 decimal places. 2
- (h) Find the area bounded by the curve $y = 3e^x$, the x axis and the lines $x = 1$ and $x = 4$. Give your answer correct to 2 decimal places. 3

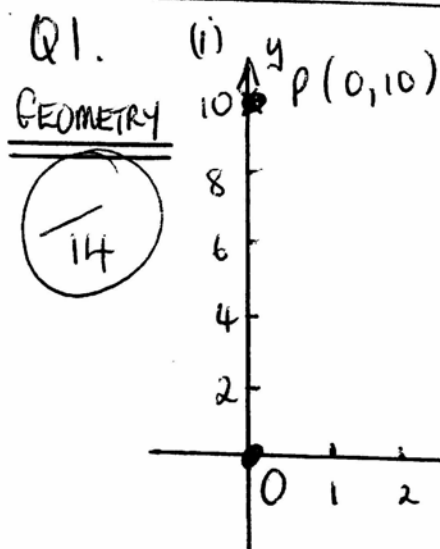
Continued over

- (i) The region bounded by the curve $y = \frac{1}{\sqrt{x+3}}$, the x axis, and the lines $x = -2$ and $x = 2$, is rotated around the x axis. Find the volume of the solid of revolution formed. Give your answer in exact form.

3

END OF EXAM

YR 12 MSC ASSESSMENT TERM 2 2003 - ANSWERS



(1) O, M, N & P labelled
- scales on both axes.

(ii) Equation of MN: $\frac{y-0}{x-5} = \frac{4-0}{8-5}$ (1)

$$\frac{y-0}{x-5} = \frac{4}{3}$$

$$3y = 4x - 20$$

$$\therefore 4x - 3y - 20 = 0 \quad (1)$$

(iii) Gradient of NP: $\frac{y_2 - y_1}{x_2 - x_1} = m$.

$$\frac{10 - 4}{0 - 8} = -\frac{6}{8} = -\frac{3}{4} \quad (1)$$

(iv) Gradient of MN: $\frac{4-0}{8-5} = \frac{4}{3}$ (1)

$$\therefore \text{gradient of NP} \times \text{gradient of MN} = -\frac{3}{4} \times \frac{4}{3} = -1 \quad (1)$$

$\therefore$ NP $\perp$ MN.

(v) MN = $\sqrt{(8-5)^2 + (4-0)^2}$ (1)

$$= \sqrt{9 + 16} = \sqrt{25}$$

$$= 5 \text{ units} \quad (1)$$

(b) $\angle ADC = 124^\circ$ (angle sum $\triangle ACD$) } ① must have 2
 $\angle FDE = 56^\circ$ (straight $\angle EDC$) } answers + 2 reasons.
 $\therefore x = 60^\circ$ (angle sum $\triangle FED$)
 ①

(c) (i) In $\triangle$'s AEM and FGC ,

① $\angle MAE = \angle FCG$ (opp. $\angle$'s, $\parallel$ gram $ABCD$)

① $MA = BF$ (M, F mid points of equal sides $\parallel$ gram $ABCD$)

① $AE = GC$ (E, G mid points of equal sides $\parallel$ gram $ABCD$)

① $\therefore \triangle AEM \equiv \triangle FGC$ (SAS)

(ii) $HG = EF$ (as $HE = GF$ from (i))

① (other facts possible - must have one - no reason necessary.)

Q2 TRIG. FUNCTIONS

14

(a) (i) $53^\circ 45' = \frac{\pi}{180} \times 53\frac{3}{4} = \frac{43\pi}{144}$ ①

(ii) $\sec \frac{3\pi}{4} = \frac{1}{\cos \frac{3\pi}{4}} = -1.4142$ ①

(b) Area = $\frac{1}{2} r^2 \theta$

$= \frac{1}{2} \times 30^2 \times \frac{\pi}{4}$ ①

$= \frac{450\pi}{4} \text{ cm}^2 = \frac{225\pi}{2} \text{ cm}^2$ ①

(c) (i) $y = \tan 6x$

$\frac{dy}{dx} = 6 \sec^2 6x$ ①

(ii) $y = \frac{x^3}{\sin x}$

$\frac{dy}{dx} = \frac{\sin x \cdot 3x^2 - x^3 \cdot \cos x}{(\sin x)^2}$ } either step ①

$\frac{dy}{dx} = \frac{x^2(3\sin x - x \cos x)}{\sin^2 x}$

(iii) $y = \frac{1}{\cos x}$

$\frac{dy}{dx} = \frac{\sin x}{\cos^2 x}$ } ① either step

$\frac{dy}{dx} = \sec x \tan x$

Q2 (continued)

$$(d) (i) \int \sin 2x \, dx$$

$$= -\cos 2x + C$$

①

$$(ii) \int_0^{\pi} (\cos \theta - 1) \, d\theta$$

$$= [\sin \theta - \theta]_0^{\pi}$$

$$= (0 - \pi) - (0 - 0)$$

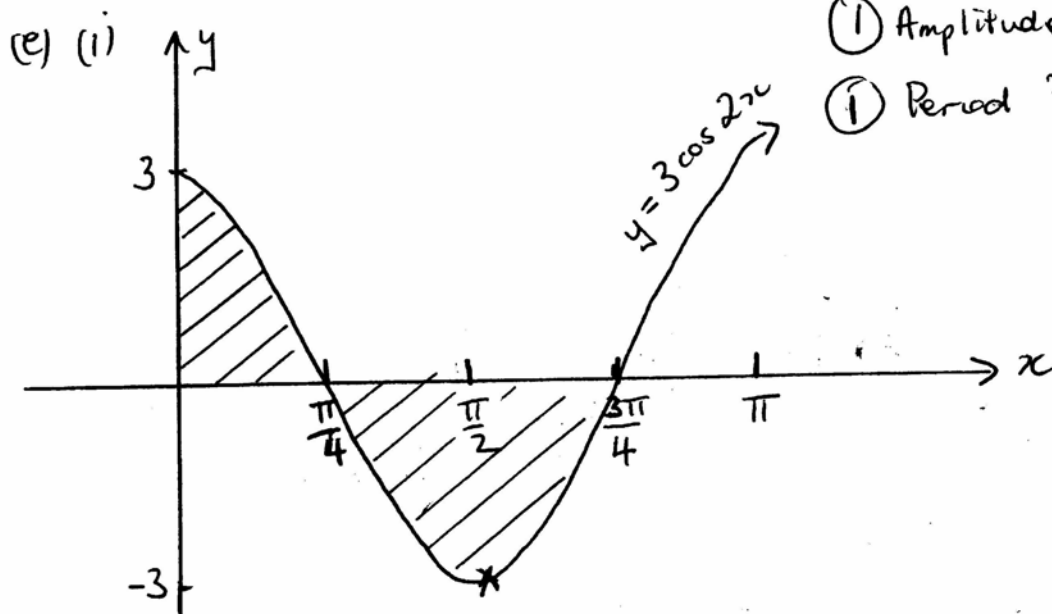
$$= -\pi$$

①

Graph must be cos graph.

① Amplitude 3

① Period π



$$(ii) \text{Area} = \int_0^{\frac{3\pi}{4}} 3 \cos 2x \, dx$$

$$= \int_0^{\frac{\pi}{4}} 3 \cos 2x \, dx + \int_{\frac{\pi}{4}}^{\frac{3\pi}{4}} 3 \cos 2x \, dx$$

$$= 3 \left[\frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{4}} + 3 \left[\frac{1}{2} \sin 2x \right]_{\frac{\pi}{4}}^{\frac{3\pi}{4}}$$

$$= 3 \left[\left(\frac{1}{2} \sin \frac{\pi}{2} - \frac{1}{2} \sin 0 \right) + \left(\frac{1}{2} \sin \frac{3\pi}{2} - \frac{1}{2} \sin \frac{\pi}{2} \right) \right]$$

$$= \dots \frac{1}{2} + 3$$

$$= 4.500 \, \text{u}^2$$

①

① for one of these steps.

QUESTION 3 (22)

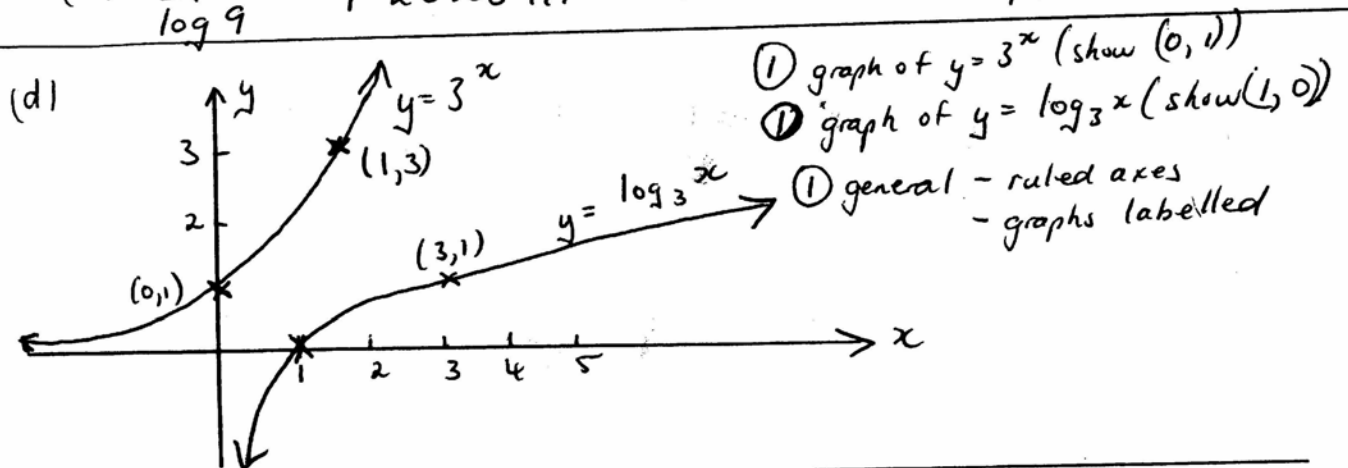
EXP. and LOG FUNCTIONS..

$$(a) (i) \frac{1}{(\sqrt[3]{125})^2} = \frac{1}{25} \quad (1) \quad (ii) (2^2)^{3x} \times 2^{4x} \div (2^3)^{2x} = 2^{6x} \times 2^{4x} \div 2^{6x} = 2^{4x} \quad (1)$$

$$(b) \frac{\log 50}{\log 5} + \frac{\log 10}{\log 5} - \frac{\log 4}{\log 5} = 3 \quad (1) \quad (3.000 \checkmark)$$

$$(c) (i) \log_6 48 = \log_6 (2^4 \times 3) = \log_6 2^4 + \log_6 3 = 4 \log_6 2 + \log_6 3 = 2.4 \quad (1)$$

$$(ii) \frac{\log 14}{\log 9} = 1.20108 \dots = 1.201 \text{ to 3 dec. pl.} \quad (1)$$



$$(e) (i) y = e^{-2x} \\ \frac{dy}{dx} = -2e^{-2x} \quad (1)$$

$$(ii) y = \frac{e}{e^x} \\ \frac{dy}{dx} = \frac{e^x}{e} (or e^{x-1}) \quad (1)$$

$$(iii) y = \log_e (5x-4) \\ \frac{dy}{dx} = \frac{5}{5x-4} \quad (1)$$

$$(iv) y = \frac{\log e^x}{3x+1} \quad \frac{dy}{dx} = \frac{(3x+1) \times 1 - x \times (3)}{(3x+1)^2} = \frac{1}{(3x+1)^2} \quad (1)$$

$$(f) (i) \int e^{2x} = \frac{1}{2} e^{2x} + C \quad (1) \quad (ii) \int \frac{4x}{2x^2+3} = \log(2x^2+3) + C \quad (1)$$

$$(g) \int_2^5 \frac{x^2}{x^3-1} = \int_2^5 \frac{1}{3} \times \frac{3x^2}{x^3-1} = \frac{1}{3} [\ln(x^3-1)]_2^5 \quad (1) \\ = \frac{1}{3} (\ln 124 - \ln 7) \\ = 0.96 \text{ to 2 dec. pl.} \quad (1)$$

③

$$(h) \quad A = \int_1^4 3e^x$$
$$= [3e^x]_1^4 \quad (1)$$

$$= 3e^4 - 3e^1$$

$$= 3(e^4 - e)$$

$$= 155.6396 \dots \quad (1)$$

$$= 155.64 \text{ u}^2 \text{ to 2 dec.pl.} \quad (1)$$

$$(i) \quad V = \pi \int_{-2}^2 y^2 dx$$
$$= \pi \int_{-2}^2 \left(\frac{1}{\sqrt{x+3}} \right)^2 dx \quad (1)$$

one of
these steps.

$$= \pi \int_{-2}^2 \frac{1}{x+3}$$

$$= \pi [\ln(x+3)]_{-2}^2 \quad (1)$$

$$= \pi [\ln 5 - \ln 1]$$

$$= \pi \ln 5 \text{ u}^3 \quad (1)$$